Secure N-Agent Ad-hoc Teamwork via Distributed Bayesian Belief Consensus and Risk-Aware Policy Optimization

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Abstract

We study *secure* N-agent ad-hoc teamwork (NAHT) in partially observable environments where a subset of *uncontrolled* teammates may be *hidden adversaries* that behave deceptively (e.g., cooperate early, sabotage near completion). Recent NAHT methods such as POAM [1] focus on generalization to diverse but *non-malicious* teammates, and can fail catastrophically under strategic sabotage. We propose a belief-to-control architecture: **(1) DBBC**, a *distributed Bayesian belief consensus* module that fuses local, uncertainty-aware adversary evidence into a calibrated shared belief map under *trusted* internal communication; and **(2) RAPO**, a *risk-aware policy optimization* method that conditions decentralized policies on this belief map and explicitly controls tail risk via a **CVaR**-constrained objective defined over belief-weighted sabotage risk. We provide a mathematically consistent formulation, practical training objectives, and an evaluation protocol on Level-Based Foraging with multiple deceptive adversaries, measuring not only mean return but also detection quality, calibration, and tail-episode robustness.

1 Introduction

Ad-hoc teamwork (AHT) requires agents to cooperate with previously unseen teammates without pre-coordination. While recent approaches leverage agent modeling and policy adaptation to handle heterogeneous teammates, they typically overlook *security*: in open systems, some teammates may be adversarial and attempt to reduce team performance via subtle, deceptive behaviors.

We consider N **strategic agents** collaborating in the presence of M **uncontrolled agents**, among which an unknown subset are **hidden adversaries** with deceptive policies. The core challenge is **distributed detection** under partial observability: a single strategic agent may not observe enough evidence to identify adversaries, so the team must *share and fuse* evidence. Beyond detection, we require **mitigation**: policies must adapt to suspected adversaries and remain robust against rare but severe sabotage events.

Contributions (proposed).

- **Secure NAHT formulation** as a Dec-POMDP with *latent agent types* (benign vs. adversarial) that are fixed per episode and must be inferred online.
- **DBBC**: an uncertainty-aware distributed Bayesian fusion rule (precision-weighted product-of-experts in logit space) that yields a *calibrated shared adversary belief map* under trusted communication.
- **RAPO**: belief-conditioned, tail-risk-sensitive policy optimization using a **CVaR**-constrained objective over a belief-defined sabotage risk functional, implemented with a Lagrangian PPO-style update.

2 Problem Formulation

2.1 Agents and hidden types

Let $\mathcal{S} = \{1, \ldots, N\}$ be the set of *strategic* agents (controlled by our algorithm) and $\mathcal{U} = \{1, \ldots, M\}$ be the set of *uncontrolled* agents (unknown policies). The full population is $\mathcal{S} \cup \mathcal{U}$, with size $N + M$.

Each uncontrolled agent $j \in \mathcal{U}$ has a latent type

$$T_j \in \mathcal{T} := \{0, 1\}, \qquad T_j = 1 \iff j \in \mathcal{U}^{\text{adv}},$$

where $\mathcal{U}^{\text{adv}} \subseteq \mathcal{U}$ is the (unknown) adversary subset. The vector $T = (T_1, \ldots, T_M)$ is sampled at the beginning of an episode and remains fixed throughout.

2.2 Dec-POMDP with latent types

We model the environment as a Dec-POMDP with an augmented latent state

$$s_t := (x_t, T) \in \mathcal{X} \times \mathcal{T}^M,$$

where $x_t \in \mathcal{X}$ is the physical environment state and T is unobserved. At time t, each strategic agent $i \in \mathcal{S}$ receives a local observation $o_t^i \in \mathcal{O}$ and selects an action $a_t^i \in \mathcal{A}$. Uncontrolled agents also take actions a_t^j according to unknown policies that depend on their type.

Let $a_t^{\mathcal{S}} = (a_t^i)_{i \in \mathcal{S}}$ and $a_t^{\mathcal{U}} = (a_t^j)_{j \in \mathcal{U}}$. The transition is

$$s_{t+1} \sim P\big(\cdot \mid s_t, a_t^{\mathcal{S}}, a_t^{\mathcal{U}}\big), \qquad a_t^{\mathcal{U}} \sim \pi_T(\cdot \mid h_t^{\mathcal{U}}),$$

where π_T denotes unknown type-dependent uncontrolled policies and $h_t^{\mathcal{U}}$ denotes uncontrolled histories (unobserved by us). From the strategic team perspective, $a_t^{\mathcal{U}}$ is latent and induces non-stationarity in observed dynamics through T.

The team receives a scalar reward

$$r_t = r(x_t, a_t^{\mathcal{S}}, a_t^{\mathcal{U}}),$$

and aims to maximize the discounted return for trajectory τ:

$$G(\tau) := \sum_{t=0}^{T-1} \gamma^t r_t, \quad \gamma \in (0, 1).$$

2.3 Information structure and trusted communication

Each strategic agent i maintains a local history

$$h_t^i = (o_0^i, a_0^i, o_1^i, a_1^i, \ldots, o_t^i),$$

and executes a decentralized policy

$$a_t^i \sim \pi_\theta(\cdot \mid h_t^i, B_t, e_i),$$

where B_t is a shared belief map over uncontrolled agents' adversariality, and e_i is an optional agent-id embedding (we use parameter sharing across strategic agents unless noted).

Threat model (TM-A). Strategic agents communicate via an internal channel that is trusted: messages exchanged among strategic agents are delivered faithfully and are not manipulated by uncontrolled agents.

Goal. Design a decentralized policy that achieves high return while controlling rare catastrophic outcomes induced by deceptive hidden adversaries, using only partial observations and trusted internal communication.

3 Proposed Method

We propose a belief-to-control architecture with two coupled modules:

1. **DBBC (Belief):** each strategic agent computes local, uncertainty-aware evidence about each uncontrolled agent's type and fuses evidence across strategic agents into a *shared calibrated belief map* B_t.
2. **RAPO (Policy):** strategic policies condition on B_t and optimize a return objective under an explicit *CVaR tail-risk constraint* on a belief-defined sabotage risk functional.

3.1 Module 1: DBBC (Distributed Bayesian Belief Consensus)

The belief module aims to estimate the posterior probability

$$b_t(j) \approx \mathbb{P}(T_j = 1 \mid \mathcal{I}_t), \quad \forall j \in \mathcal{U},$$

where $\mathcal{I}_t$ denotes the union of strategic agents' information up to time t (histories plus trusted messages).

3.1.1 Local evidence modeling (per agent i)

Each strategic agent encodes its history with a recurrent encoder:

$$z_t^i = \mathrm{Enc}_\phi(h_t^i), \quad \mathrm{Enc}_\phi : (\mathcal{O} \times \mathcal{A})^{t+1} \to \mathbb{R}^d.$$

For each uncontrolled agent $j \in \mathcal{U}$, agent i builds relational features ψ_t^{ij} (e.g., relative position/velocity if observable, co-targeting the same food, blocking interactions, last-seen time, etc.). A neural head outputs a Gaussian distribution over a latent *adversariality logit*:

$$\ell_{ij,t} \mid (z_t^i, \psi_t^{ij}) \sim \mathcal{N}(\mu_{ij,t}, \sigma_{ij,t}^2), \tag{1}$$

$$(\mu_{ij,t}, \sigma_{ij,t}) = f_\phi(z_t^i, \psi_t^{ij}), \quad \sigma_{ij,t} > 0. \tag{2}$$

Local belief. We use a plug-in approximation

$$b_{ij,t} := \mathrm{sigm}(\mu_{ij,t}), \qquad \mathrm{sigm}(x) = \frac{1}{1+e^{-x}},$$

while $\sigma_{ij,t}$ represents uncertainty that will be used for fusion.

3.1.2 Trusted Bayesian fusion (precision-weighted product-of-experts)

Under TM-A, agents share $(\mu_{ij,t}, \sigma_{ij,t})$ for all $j \in \mathcal{U}$. Assuming agents provide conditionally independent Gaussian "measurements" of the same latent logit for target j, the product of Gaussian experts yields another Gaussian whose mean is a precision-weighted average.

Define the precision weight

$$w_{kj,t} := \frac{1}{\sigma_{kj,t}^2 + \varepsilon}, \qquad \varepsilon > 0.$$

The fused logit mean and variance are

$$\bar{\mu}_{j,t} = \frac{\sum_{k\in\mathcal{S}} w_{kj,t}\,\mu_{kj,t}}{\sum_{k\in\mathcal{S}} w_{kj,t}}, \tag{3}$$

$$\bar{\sigma}^2_{j,t} = \left(\sum_{k\in\mathcal{S}} w_{kj,t}\right)^{-1}. \tag{4}$$

3.1.3 Shared belief map

The shared adversary belief map is

$$b_t(j) := \mathrm{sigm}(\bar{\mu}_{j,t})\,, \quad B_t := \{b_t(j)\}_{j\in\mathcal{U}}. \tag{5}$$

Optionally, policies can also condition on $\bar{\sigma}_{j,t}$ to act conservatively when uncertainty is high.

3.1.4 Training DBBC (detection and calibration)

During training in simulation, we know ground-truth types $y_j := T_j \in \{0,1\}$. We train the belief module using detection and calibration losses:

Detection loss.

$$\mathcal{L}_{\mathrm{det}} = \sum_t \sum_{j\in\mathcal{U}} \mathrm{BCE}(b_t(j),\, y_j)\,. \tag{6}$$

Calibration loss.

$$\mathcal{L}_{\mathrm{cal}} = \sum_t \sum_{j\in\mathcal{U}} (b_t(j) - y_j)^2\,. \tag{7}$$

Total belief loss.

$$\mathcal{L}_{\mathrm{belief}} = \mathcal{L}_{\mathrm{det}} + \lambda_{\mathrm{cal}}\mathcal{L}_{\mathrm{cal}}, \qquad \lambda_{\mathrm{cal}} > 0 \text{ small}. \tag{8}$$

3.2 Module 2: RAPO (Risk-Aware Policy Optimization)

RAPO optimizes strategic policies for robustness against deceptive adversaries by explicitly controlling *tail risk*.

3.2.1 Belief-conditioned decentralized policy

Each strategic agent executes a decentralized policy conditioned on its history and the shared belief map:

$$a^i_t \sim \pi_\theta(\cdot \mid h^i_t, B_t, e_i). \tag{9}$$

The belief map enables mitigation by reducing reliance on agents with high $b_t(j)$ and by avoiding risky interaction patterns.

3.2.2 Belief-defined sabotage risk

We define an interaction event indicator $\kappa_t(i,j) \in \{0,1\}$ for whether strategic agent i is meaningfully entangled with uncontrolled agent j at time t. In LBF, natural choices include (examples): (i) adjacency while attempting to cooperatively carry the same food item, (ii) co-targeting the same food (competing near the item), (iii) blocking events when i is carrying and j occupies a critical adjacent cell.

Abstractly, we treat $\kappa_t(i,j)$ as a task-dependent binary predicate computed from observable features (or from simulator events in training):

$$\kappa_t(i,j) = \mathbb{I}(i \text{ and } j \text{ are in a high-impact interaction at time } t).$$

We define per-agent instantaneous risk as belief-weighted interaction mass:

$$\rho_t^i := \sum_{j \in \mathcal{U}} b_t(j)\, \kappa_t(i,j), \tag{10}$$

team risk:

$$\rho_t := \sum_{i \in \mathcal{S}} \rho_t^i, \tag{11}$$

and the discounted trajectory risk:

$$R(\tau) := \sum_{t=0}^{T-1} \gamma^t \rho_t. \tag{12}$$

This risk is *structurally grounded* in the posterior belief produced by DBBC, making it suitable for calibrated tail-risk control.

3.2.3 CVaR (upper-tail) and optimization-friendly form

Let R be the random variable $R(\tau)$ induced by rolling out π_θ. We focus on *upper-tail* risk. For a confidence level $\beta \in (0,1)$ (e.g., $\beta = 0.9, 0.95$), define

$$\mathrm{VaR}_\beta(R) := \inf\{\eta \in \mathbb{R} : \mathbb{P}(R \le \eta) \ge \beta\},$$

and the corresponding *upper-tail* conditional value-at-risk:

$$\mathrm{CVaR}_\beta(R) := \mathbb{E}[R \mid R \ge \mathrm{VaR}_\beta(R)]\,.$$

Thus $\mathrm{CVaR}_{0.95}$ measures expected risk among the worst 5% trajectories.

We use the Rockafellar–Uryasev representation:

$$\mathrm{CVaR}_\beta(R) = \min_{\eta \in \mathbb{R}} \left[\eta + \frac{1}{1-\beta} \mathbb{E}\big[(R-\eta)_+\big] \right], \quad (x)_+ := \max(x,0). \tag{13}$$

3.2.4 Risk-constrained objective

We aim to maximize expected return while constraining tail risk:

$$\max_\theta \ \mathbb{E}\big[G(\tau)\big] \quad \text{s.t.} \quad \mathrm{CVaR}_\beta\big(R(\tau)\big) \le \delta, \tag{14}$$

where δ is a user-specified risk budget and β determines the tail severity.

3.2.5 Lagrangian PPO update (RAPO)

We solve (14) via a Lagrangian relaxation with dual variable $\lambda \ge 0$:

$$\max_\theta \ \min_{\lambda \ge 0} \ \mathcal{L}(\theta,\lambda) = \mathbb{E}\big[G(\tau)\big] - \lambda\left(\mathrm{CVaR}_\beta(R(\tau)) - \delta\right). \tag{15}$$

Algorithm 1 Training DBBC + RAPO (high-level)

Initialize belief parameters ϕ, policy parameters θ, dual variable $\lambda \geq 0$
for iteration $= 1, 2, \ldots$ **do**
  Collect on-policy trajectories $\{\tau_k\}_{k=1}^{B}$ using π_θ with DBBC producing B_t
  Belief update: minimize $\mathcal{L}_{\text{belief}}$ in (8) w.r.t. ϕ
  Risk/return computation: compute G_k and R_k using (12)
  Estimate $\widehat{\text{CVaR}}_\beta(R)$ via (13) (sample approximation)
  Policy update (PPO-style): update θ to increase $\widehat{\mathcal{L}}(\theta, \lambda)$ from (15)
  Dual update: $\lambda \leftarrow \max(0,\ \lambda + \eta_\lambda(\widehat{\text{CVaR}}_\beta(R) - \delta))$
end for

Sample-based CVaR estimation. From an on-policy batch $\{\tau_k\}_{k=1}^{B}$, compute $R_k := R(\tau_k)$ and $G_k := G(\tau_k)$. Using (13), estimate

$$\widehat{\text{CVaR}}_\beta(R) \approx \min_\eta \left[\eta + \frac{1}{(1-\beta)B} \sum_{k=1}^{B} (R_k - \eta)_+ \right].$$

Then update θ with PPO-style clipped gradients on a combined objective where the return advantage is encouraged while tail risk is penalized through the CVaR term. The dual variable is updated by projected ascent:

$$\lambda \leftarrow \max\left(0,\ \lambda + \eta_\lambda \left(\widehat{\text{CVaR}}_\beta(R) - \delta\right)\right).$$

4 Overall Training Procedure

We train DBBC and RAPO in a CTDE manner (centralized training with decentralized execution). Execution remains decentralized since each agent conditions only on (h_t^i, B_t, e_i) and trusted messages are used only to compute B_t.

Practical recipe. We pretrain DBBC for several iterations using supervised type labels in simulation, then jointly fine-tune with RAPO so the belief map remains calibrated and behaviorally relevant.

5 Experimental Design

5.1 Environment: Level-Based Foraging (LBF) with deceptive adversaries

We evaluate on LBF with larger maps (e.g., 10×10 and 15×15) to emphasize partial observability and distributed evidence. We introduce adversarial behaviors among uncontrolled agents:

- **Multiple adversaries:** $N = 3$ strategic vs. $M = 3$ uncontrolled with $K \in \{1, 2\}$ adversaries, where the identities are hidden and K is unknown to strategic agents.
- **Deceptive impostors:** adversaries imitate cooperative foraging early, then sabotage near completion (e.g., blocking when strategic agents carry, competing for the same final food, "hovering" to cause delays, or disrupting cooperative carrying events).
- **Scalability:** vary (N, M, K), e.g., $N \in \{2, 3, 5\}$, $M \in \{2, 3, 5\}$, and $K \in \{1, 2, 3\}$.

5.2 Baselines

1. **Vanilla NAHT/AHT (POAM-like) without adversary belief and risk control.**
2. **Independent belief (I-BAE):** each agent estimates $b_{ij,t}$ locally; no fusion across strategic agents.
3. **Belief-only:** DBBC beliefs are learned and fused, but policies are trained with mean-return PPO (no CVaR constraint).
4. **Risk-only:** PPO with a hand-crafted cost (using κ) and a Lagrangian penalty, but without learned adversary belief (no B_t).
5. **Oracle upper bound:** strategic agents observe T (adversary identities) from the start.

5.3 Metrics

We report metrics beyond average reward to demonstrate security and robustness:

- **Task performance:** mean return, completion rate, efficiency (steps/food).
- **Detection:** AUC-ROC/F1 for adversary identification, detection delay (time-to-identify).
- **Calibration:** Brier score and ECE (reported).
- **Tail robustness:** completion rate and mean return on the worst $(1-\beta)$ fraction of episodes (aligned with CVaR).
- **Risk:** empirical $\widehat{\mathrm{CVaR}}_\beta(R)$ and frequency of violating the risk budget δ.

5.4 Ablations

- Remove fusion: I-BAE vs. DBBC.
- Remove uncertainty: fusion with constant σ vs. precision-weighted fusion.
- Remove risk control: mean-return PPO vs. RAPO (CVaR constraint).
- Corrupt or randomize B_t during training/testing to measure reliance and robustness to belief quality.

6 Conclusion

We propose a secure NAHT framework that explicitly addresses hidden deceptive adversaries under partial observability with trusted internal communication. DBBC provides calibrated, uncertainty-aware adversary belief consensus across strategic agents, while RAPO optimizes belief-conditioned policies with explicit tail-risk control via a CVaR constraint. The combination targets both distributed detection and robust mitigation, aiming to substantially improve worst-case safety and performance compared to existing NAHT baselines.

References

[1] Caroline Wang, Arrasy Rahman, Ishan Durugkar, Elad Liebman, and Peter Stone. N-Agent Ad Hoc Teamwork. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. `https://proceedings.neurips.cc/paper_files/paper/2024/file/cabf611498431ad89a85ace75f790d93-Paper-Conference.pdf`.